



E-CELL

DOMAIN

T A S K 3



P R O B L E M S T A T E M E N T

Enterprise CRM platforms like Salesforce and HubSpot manage thousands of customer interactions daily, yet smaller organizations and startups lack access to intelligent, AI-native operational tools. Your task is to build an AI-integrated CRM platform for a fictional company founded by the E-Cell heads, where the system manages customers, customer logs, support tickets, interaction history, and operational workflows similar to enterprise-grade platforms.

The platform must integrate LLM-powered features including ticket summarization, customer interaction memory, chat history management, AI agents using LangChain and LangGraph, and end-to-end workflow orchestration. Teams must also design a cohort analysis engine and a HEART framework evaluation dashboard to demonstrate measurable business impact.

D A T A S E T

Synthetic customer and ticket datasets must be generated using LLMs. Your dataset must include a minimum of 500 customer profiles, 1,000 support tickets spanning multiple categories, interaction logs covering at least six months, and per-customer behavioral event sequences. All data must be ingested into the live CRM system and demonstrated at evaluation time.

C R M E v a l u a t i o n S e t u p

Using your deployed system, establish a clear evaluation framework. Measurable performance metrics include:

- HEART Framework Scores (Happiness, Engagement, Adoption, Retention, Task Success) computed from live CRM data
- Cohort Retention Rate and Churn Prediction Accuracy per customer segment
- LLM Agent Response Quality, Hallucination Rate, and Ticket Resolution Latency
- Any other explicit operational KPI derivable from standard CRM analytics frameworks

⚠ IMPORTANT NOTE: Your HEART framework analysis, AI agent behavior definitions, and cohort segmentation logic must be clearly defined and justified in your System Report. Vague feature implementations, missing retention curves, or unhandled LLM hallucinations in agent responses will negatively impact your evaluation score.



What You Need to Build

Your project must implement a five-module architecture. Each module is independently scoped but tightly integrated. Below are the first two core modules in detail.

Module 1: Customer & Ticket Management

- Design and implement a customer database with full CRUD operations, interaction history tracking, and configurable behavioral metadata fields.
- Build a support ticket system with full lifecycle state management: Open, In Progress, Escalated, Resolved, and Closed.
- Implement customer segmentation by industry, tenure, ticket frequency, and engagement score using configurable rule logic.
- Create per-customer timeline views showing all interaction events (tickets, calls, chats) sorted chronologically across channels.
- Support bulk data ingestion from the LLM-generated synthetic dataset on system startup with validation and deduplication.

Module 2: LLM-Powered Intelligence Layer

Convert raw ticket and interaction data into intelligent, AI-assisted operations using one or more of the following components:

Ticket Summarization Engine: LangChain summarization chains Output: Key issues, urgency, resolution path > Context-aware multi-turn thread compression > Configurable summary length and tone 01	AI Agent Workflows Engine: LangGraph statemachines Format: Multi-step resolution pipelines > Automated ticket routing by category & priority > Agent escalation and handoff state transitions 02
Interaction Memory Setup: Per-customer persistent memory buffers Format: Short-term + long-term context layers Cross-session chat history retrieval > Memory-aware response generation > 03	Response Generation (Optional) Usage: Draft replies for human agents Setup: Tone rules, formatting templates > Hallucination-guarded output with source citation 04

Module 3: Cohort Analytics



- Segment customers into cohorts by acquisition date, industry vertical, product tier, or any configurable behavioral tag applied during onboarding.
- Track and visualize cohort retention curves, churn windows, re-engagement rates, and time-to-first-resolution metrics per cohort.
- Compute resolution time metrics segmented by cohort, ticket category, and assigned agent group for SLA monitoring.
- Integrate predictive churn scoring using engagement signals, ticket frequency, recency of last interaction, and session depth.
- Export cohort analysis reports in structured JSON and human-readable PDF formats for stakeholder and investor review.

Option A (Local LLM): Implement Ollama / vLLM (Llama-3-8B-Instruct) to run all AI inference fully on-premise, ensuring total data privacy, zero external API cost, and sub-second response latency for bulk cohort operations.

Option B (API LLM): Integrate Google Gemini API (gemini-1.5-flash) using its zero-cost developer tier to leverage state-of-the-art reasoning, expansive context windows, and superior cross-lingual customer interaction support.

Module 4: HEART Framework Dashboard

Display all five Google HEART dimensions with live calculation from your CRM data. Each metric must be backed by a clearly defined signal source and computation methodology in the System Report:

- **H – HAPPINESS:** Customer satisfaction scores (CSAT) per cohort and interaction channel; average NPS per ticket category and support agent.
- **E – ENGAGEMENT:** Active customers per week, ticket open rates, response depth per thread, and average session duration in the CRM portal.
- **A – ADOPTION:** Feature adoption rate across CRM modules, onboarding completion rate, and percentage of customers using AI-assisted interactions.
- **R – RETENTION:** Monthly retention curves, cohort survival analysis, per-customer churn probability flags, and average customer lifespan by segment.
- **T – TASK SUCCESS:** Ticket resolution rate, AI-assisted vs human completion comparison, escalation frequency, and first-contact resolution percentage.

Module 5: Backend API Architecture



- Serialize and persist all CRM state: customer profiles, ticket index, cohort configurations, agent memory snapshots, and system prompt templates.
- Deploy a production-grade FastAPI application with OpenAPI auto-documentation accessible at **/docs** and versioned routing under **/api/v1/**.
- Implement role-based access control with at least four distinct roles: Agent, Supervisor, Admin, and Analytics - read only.
- All endpoints must return structured JSON with verifiable audit metadata: timestamps, agent ID, source confidence scores, and processing latency.

DEPLOYMENT ENDPOINTS:

POST /customers	Input: {customer_data} Output: {id, status, cohort_assignment}
POST /tickets/create	Input: {customer_id, title, description, priority} Output: {ticket_id, category, assigned_agent}
POST /tickets/{id}/summarize	Input: {ticket_id} Output: {summary, key_issues, suggested_response}
POST /query/agent	Input: {customer_id, query} Output: {answer, source, confidence, agent_id}
GET /cohorts/analysis	Output: {cohort_id, retention_curve, churn_rate, heart_scores}

Stage 6: System Evaluation

Evaluate the full CRM pipeline and produce a clear performance comparison report. Required Metrics:

- HEART SCORES:** All five dimensions computed from live CRM data, with per-cohort breakdown and trend charts.
- AGENT QUALITY:** LLM response faithfulness, hallucination rate, and answer relevance per ticket category and agent variant.
- SYSTEM LATENCY:** API response time per endpoint under 50, 200, and 500 concurrent operations.
- COHORT ACCURACY:** Churn prediction precision/recall, retention curve smoothness, and cohort coverage completeness.
- RESOLUTION RATE:** Ticket closure rate, AI-assisted vs human resolution ratio, and escalation frequency trend.



Minimum Requirements

- Functional customer and ticket management system with full CRUD and lifecycle state transitions running
- LLM-powered ticket summarization with at least one working LangChain summarization chain deployed
- At least one operational LangGraph agent workflow for ticket routing or automated response generation
- Persistent customer interaction memory with minimum short-term context window support per customer
- Cohort analysis engine with retention curve computation and basic churn probability scoring
- HEART framework metric calculations with live dashboard visualization covering all five dimensions
- Synthetic dataset of 500+ customer profiles and 1,000+ tickets generated via LLM and fully ingested
- Production FastAPI backend with Swagger documentation auto-accessible at /docs
- Role-based access control implemented for at least two distinct user roles
- Working query interface with structured JSON inputs and outputs for all five core API endpoints

Deliverables

1 • Source Code

A highly organized GitHub repository with clean, modular, production-ready Python and frontend files. Each module (CRM, agents, cohort, HEART, API) should be independently runnable from its own script. Avoid monolithic notebooks for final submissions.

2 • System Report

A concise written report (PDF or Markdown) covering: module architecture decisions and LLM selection rationale; agent prompt design and LangGraph state machine documentation; cohort segmentation methodology, retention curve analysis, and churn model evaluation; HEART framework metric definitions with data-driven commentary and benchmark charts.

3 • Live Demo

Be ready to demonstrate the following during evaluation:

- Live ticket ingestion, LLM summarization output, and LangGraph agent resolution pipeline in action
- HEART framework dashboard with live cohort metrics, retention curves, and churn flags
- FastAPI /query/agent endpoint resolving a complex multi-turn customer situational prompt
- Cohort analysis report export showing clear metric tracing and churn prediction scores

Suggested Folder Structure



```
project/
├── data/                #synthetic customer/ticket dataset and CRM records
├── notebooks/          #exploratory LLM prompt testing and cohort prototyping
├── src/
│   ├── crm.py ───      #customer and ticket CRUD and lifecycle management
│   ├── agents.py ───   #LangGraph workflows and LangChain chain definitions
│   ├── memory.py ───   #per-customer interaction memory and chat history
│   ├── cohort.py ───   #cohort segmentation, retention curves, churn scoring
│   └── heart.py ───    #HEART framework metric computation and aggregation
├── api/
├── app.py ───          #FastAPI backend with all CRM endpoint definitions
├── dashboard/         #frontend HEART dashboard and cohort visualization UI
├── models/            #saved LLM configs, agent states, and prompt templates
├── README.md          #comprehensive system engineering and evaluation report
└── requirements.txt    #environment dependency declarations
```



S U B M I S S I O N G U I D E L I N E S

DEADLINE – 5TH JULY

Meet the deadlines, late submissions will not be entertained. PDF format of the ppt must be submitted.

For any clarifications reach out to:

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